AI SALES AGENT

Deployment architecture

How the Phase 1 system is deployed on AWS, including network boundaries, data flow, and deferred items.

Phase 1 MVP · 21 October 2026 · Prepared from the canonical documents in docs/

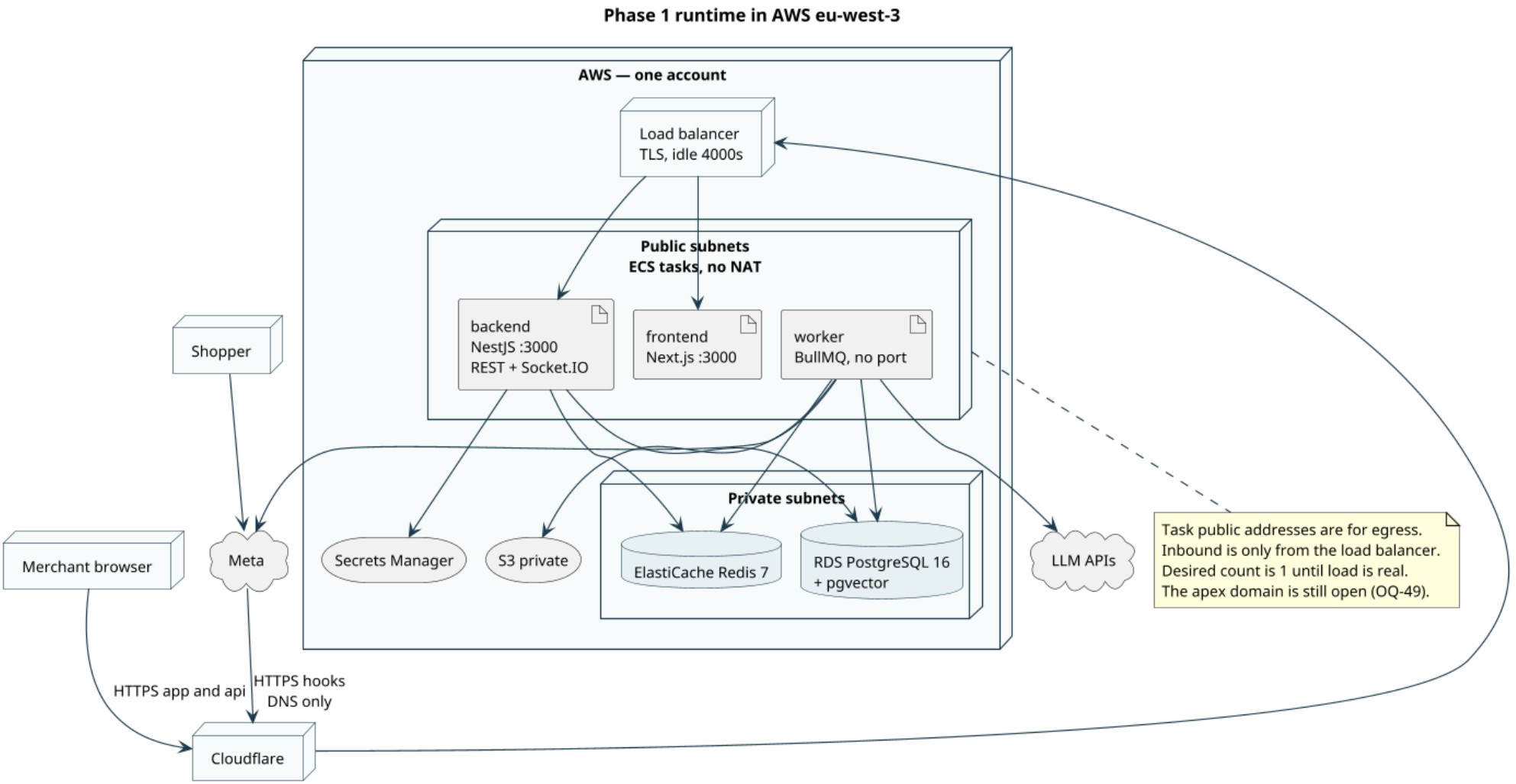
Scope of this document

This document describes how the Phase 1 system is hosted. The topology, instance sizes, and security boundaries are taken from the approved deployment specification. Nothing in the diagram is a proposed alternative. Items the specification explicitly postpones are labelled as deferred, not drawn as if they existed.

The region is eu-west-3 (Paris), with eu-central-1 only if a required SKU is missing. There is one AWS account. Names are prefixed for staging and production. Infrastructure is Terraform, one root module, with a separate variable file per environment.

Deployment diagram

Runtime nodes for one environment. Staging is a smaller copy of this shape, not a second target group on production. Source: `submission/sources/diagram-source-files/14-deployment.puml`.



Nodes and what they do

Node	Role
Merchant browser	Opens the command center on app and the API and Socket.IO endpoint on api.
Shopper and Meta	The shopper stays in Instagram or Messenger. Meta calls only the hooks host.
Cloudflare	Proxied DNS for app and api. Hooks is DNS-only so Meta is not challenged. TLS is Full (strict), minimum TLS 1.2. WebSockets are enabled.
Application load balancer	One per environment. ACM certificate for app, api, and hooks. Idle timeout 4000 seconds so Socket.IO survives. Stickiness is off. Port 80 redirects to 443.
frontend	ECS Fargate, Next.js standalone, port 3000. Health check is GET /health. 256 CPU, 512 MB. Desired count 1.
backend	ECS Fargate, NestJS, port 3000. REST and Socket.IO. Readiness is GET /ready, which checks PostgreSQL and Redis. Production is 512 CPU and 1024 MB.
worker	Same image, command node dist/worker, no listener. Stop timeout 120 seconds so the current job can finish. Production is 512 CPU and 1024 MB.
migrate task	A one-shot Fargate task running prisma migrate deploy. Migrations do not run inside the API process.
RDS PostgreSQL 16	Private. pgvector. Production is db.t4g.small with 50 GB. Daily backups and point-in-time recovery for 7 days.
ElastiCache Redis 7	Private, one node, cluster mode off, eviction noeviction. Snapshots off: Redis is not the system of record.
S3	Private, SSE-S3, versioning on. Images are served with signed GET. There is no CloudFront.
Secrets Manager	Application secrets and provider tokens. GitHub uses OIDC. There are no long-lived AWS keys in GitHub.
CloudWatch	Logs and alarms for API errors, worker failures, AI latency, and Meta failures.
LLM providers	Called by the worker over HTTPS. They are outside the VPC.

Network

The VPC is 10.0.0.0/16 with two availability zones, because the load balancer needs two. There is an internet gateway and no NAT gateway. Public subnets are 10.0.0.0/20 and 10.0.16.0/20 and hold the load balancer and the ECS tasks. Private subnets are 10.0.32.0/20 and 10.0.48.0/20 and hold RDS and Redis. Tasks have public addresses so they can reach Meta and the model providers without NAT. Those addresses are not an open application port: security groups allow port 3000 only from the load balancer. The worker has no inbound rule. RDS accepts 5432 only from the backend, the worker, and the migrate task. Redis accepts 6379 only from the backend and the worker. The only VPC endpoint is the free S3 gateway.

Path	Protocol
Browser to Cloudflare to the load balancer	HTTPS. WebSocket upgrade on api.
Meta to hooks	HTTPS POST /api/v1/webhooks/meta and /deauthorize. Signature is checked before any row is written.
Load balancer to tasks	HTTP on port 3000 inside the VPC.
Tasks to RDS and Redis	TCP 5432 and TCP 6379.
Worker to Meta and to model providers	HTTPS.
Tasks to S3 and Secrets Manager	HTTPS.

Runtime data flow

A customer message arrives at hooks. The backend verifies the signature, stores the raw event, enqueues mapping, and returns 200. The worker maps that event to a canonical message, the conversation module stores it, and a second job runs the model turn under a Redis lock. A reply in auto-send mode is queued and delivered by the integrations module. The inbox hears about the message through Socket.IO, using Redis to fan out across backend tasks. If Redis is lost, jobs can be rebuilt from the database and from Meta's retry. PostgreSQL remains the record.

Uploads go through the API. The frontend cannot reach S3, RDS, or Meta. CORS allows only the app origin.

Environments

	Local	Staging	Production
Run	Docker Compose: api, worker, web, Postgres, Redis, MinIO	Fargate, desired count 1	Fargate, desired count 1
Data	Containers	RDS db.t4g.micro, 20 GB; Redis cache.t4g.micro	RDS db.t4g.small, 50 GB; Redis cache.t4g.micro
Secrets	A gitignored env file	Secrets Manager prefix for staging	Secrets Manager prefix for production
Release	docker compose up	Every push to main	A reviewed tag v*
Meta	Test app or a tunnel	Test app on the staging hooks host	Live app on the production hooks host

Local services use the same environment variable names as Fargate. Local processes are not pointed at the staging database.

Release behaviour

Images are aisales-web and aisales-api. The backend, the worker, and the migrate task of one release share one image digest. The pipeline builds, pushes, applies Terraform for staging when infrastructure changed, migrates, and rolls ECS. A failed smoke test rolls the three services back to the previous task definition and does not reverse the migration. A failed migration does not roll the services; it is fixed forward. Auto-scaling is off. When load is real, the worker scales on queue depth or oldest job age, and the backend scales on socket connections and latency, each from 1 to 3 tasks. The frontend stays at 1 until it is the constraint.

Recovery point objective is at most 5 minutes, from RDS point-in-time recovery. Recovery time objective is at most 4 hours. A restore drill is required before the first paying tenant and then monthly. Redis is not backed up. S3 versioning covers object history.

Deferred, and not drawn as current design

- Tasks in private subnets, a NAT gateway, multi-AZ RDS, and RDS Proxy are documented as later work. They are not part of this deployment.
- Kubernetes, a fourth realtime service, AWS WAF, CloudFront, and a second account are forbidden for the MVP.
- The apex domain name is OQ-49. The three host roles are fixed. The diagram labels them as app, api, and hooks without inventing a domain.
- Auto-scaling bounds above are the documented trigger for later load. Desired count today is 1.